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Explainer: A Geometric Profile of Semantic Information in Text

A self-contained, technical-but-gentle walkthrough of the paper. Every symbol, every concept, defined before it is used. Written to be read in order, from the top.

1. The problem the paper solves

Question. Given a piece of text T , how much “meaning” does it carry — and how is that meaning structured?

Why existing tools don’t answer it.

- **Shannon entropy** measures uncertainty over symbols (characters, tokens). By design, it is **indifferent to meaning**. A random shuffle of words has roughly the same entropy as coherent prose.
- **Pairwise similarity metrics** (BERTScore, BLEURT, ROUGE) compare *two* texts. They can’t characterize the richness of *one* text on its own.
- **Philosophical accounts** (Bar-Hillel & Carnap, Dretske, Floridi) define semantic information via logic or situation theory — rigorous, but not computable on real text.

The paper’s approach. Map each sentence of T to a high-dimensional vector using a neural sentence-embedding model. The **geometry of the resulting point cloud** — where the points sit, how they spread, how they connect — encodes the text’s semantic structure. Measure meaning by measuring geometry.

2. Primitives

Every term the paper uses, defined from the ground up.

2.1 Sentence embedding

A **sentence-embedding model** E is a neural network that maps a piece of text T to a vector $E(T) \in \mathbb{R}^n$. Typical dimension $n = 768$ (for `all-mpnet-base-v2`, the paper’s primary model). Outputs are ℓ_2 -normalized (unit length). The design goal of E : semantically similar sentences get nearby vectors; dissimilar sentences get distant vectors.

2.2 Segments

A text T is split into segments — typically sentences — giving $T = (T_1, T_2, \dots, T_k)$. Each segment gets embedded: $e_i = E(T_i) \in \mathbb{R}^n$. The raw embedding cloud is $X_T = \{e_1, \dots, e_k\}$.

- k = number of segments in the text.
- n = embedding dimension (e.g., 768).

2.3 Mean and covariance

From the cloud X_T : - $\mu_T \in \mathbb{R}^n$ — the mean vector (centroid): where the text sits in embedding space. - C_T — the covariance matrix of the segments around μ_T . An $n \times n$ symmetric positive-semidefinite matrix describing how the segments scatter.

2.4 Neutral baseline

A **baseline corpus** of semantically generic sentences (e.g., “The meeting is scheduled for tomorrow afternoon.”). Embedding these gives: - $\mu_0 \in \mathbb{R}^n$ — mean of generic discourse; the “semantic origin.” - Σ_0 — covariance of generic discourse; the “natural spread” of ordinary English.

Displacement of a text’s centroid from μ_0 measures how far the text is from generic prose.

2.5 Mahalanobis distance

Ordinary Euclidean distance treats all directions equally. **Mahalanobis distance** is direction-aware: given covariance Σ_0 , the Mahalanobis norm of a vector v is

$$\|v\|_M = \sqrt{v^\top \Sigma_0^{-1} v}.$$

Intuition: directions in which the baseline varies widely (common drift) are **down-weighted**; directions in which it’s narrow (unusual for generic prose) are **up-weighted**. Moving one inch into a direction generic English never occupies counts more than one inch along a direction it wobbles in all the time.

2.6 Ledoit–Wolf shrinkage

In 768-dimensional space with a finite baseline corpus, the sample covariance $\hat{\Sigma}_0$ is near-singular — inverting it gives garbage. **Ledoit–Wolf shrinkage** mixes the sample covariance with a scaled identity matrix λI :

$$\Sigma_{LW} = (1 - \alpha) \cdot \hat{\Sigma}_0 + \alpha \cdot \lambda I,$$

with α chosen analytically to minimize expected squared error. The result is a well-conditioned estimate that can be inverted safely. The paper uses this whenever Σ_0^{-1} is needed.

2.7 Rank of a covariance

rank(C) of an $n \times n$ matrix = the number of nonzero eigenvalues, i.e., the dimension of the subspace that C actually occupies. If all segments lie on a line, **rank(C) = 1**. If they span a plane, **rank = 2**. Geometrically: the **number of independent directions** of variation.

2.8 Deduplication and the threshold τ

To prevent exact or near-duplicates from inflating measurements, the framework runs **agglomerative clustering** on X_T with cosine distance, merging any two segments whose cosine similarity exceeds a threshold $\tau \in (0, 1)$. Each cluster is replaced by its (renormalized) centroid. The result is a **deduplicated set**:

$$\tilde{T} = \{c_1, \dots, c_m\}, \quad m \leq k.$$

- **High τ (e.g., 0.90)** — strict. Only near-identical sentences merge. Fine resolution.
- **Low τ (e.g., 0.70)** — lenient. Topically related sentences merge. Coarse resolution. Default for natural prose.

2.9 Effective rank (smooth rank)

Raw **rank(C)** is brittle: in high-dimensional embeddings, noise makes nearly every eigenvalue strictly nonzero, so rank saturates at $\min(k-1, n)$ and stops discriminating. The **effective rank** is a smooth replacement:

$$D_{\text{eff}} = \exp(H(p)), \quad p_i = \lambda_i / \sum_j \lambda_j,$$

where λ_i are the eigenvalues of C and $H(p) = -\sum p_i \log p_i$ is their Shannon entropy. Intuition:

- All variance on one axis $\rightarrow D_{\text{eff}} = 1$.
- Variance equally spread over k axes $\rightarrow D_{\text{eff}} = k$.
- Uneven spread $\rightarrow$ something in between, rewarding directions carrying real mass and discounting noise-level directions.

A **continuous idea-count**: how many genuine dimensions of variation the cloud spans.

3. The three-coordinate profile $P_E(T) = (N, B, I)$

Instead of collapsing the geometry of X_T into one number, the paper reports **three coordinates** capturing different facets of “semantic information.”

3.1 Novelty **N**

$$S_M(T) = \|\mu_T - \mu_0\|_M = \sqrt{(\mu_T - \mu_0)^\top \Sigma_0^{-1} (\mu_T - \mu_0)}$$

$$N(T) = \log(1 + S_M(T)).$$

In words. Measure how far the text’s centroid sits from the baseline, in Mahalanobis units (baseline-variance-aware), then squash with $\log(1 + \cdot)$ so outliers don’t dominate.

- **N** = 0 $\Rightarrow$ text indistinguishable from generic prose.
- **N** small $\Rightarrow$ ordinary prose.
- **N** large $\Rightarrow$ semantically distant from generic discourse (unusual register, specialized domain).

The answer to: “*How unusual is this text?*”

3.2 Breadth **B**

Computed on the **deduplicated** set $\tilde{\mathbf{T}}$, not the raw cloud $\mathbf{X}_{\mathbf{T}}$. This enforces that duplicates can’t inflate it.

$$R(\tilde{T}) = \frac{1}{m} \sum_{j=1}^m [1 - \cos(c_j, \mu_{\tilde{T}})]$$

$$B(T) = D_{\text{eff}}(\tilde{T}) \cdot R(\tilde{T}).$$

Two factors multiplied: - **D_eff** — effective rank of the centroid cloud’s covariance. *How many independent directions* the ideas span. - **R** — mean cosine distance of each centroid to the centroid-cloud’s mean. *How far apart* the ideas sit within those directions.

Multiplication ensures that genuine breadth requires **both** many directions **and** meaningful spread. The answer to: “*How many genuinely different ideas does this text cover?*”

3.3 Integration **I**

$$I_{2\text{-NN}}(T) = \frac{1}{m} \sum_j \text{secondmax}_{l \neq j} \cos(c_j, c_l).$$

For each centroid c_j , look at its **second-nearest** neighbor (second-most-similar centroid). Average across all centroids.

Why second-nearest rather than nearest? A single near-duplicate pair that survived deduplication would fake 1-NN coherence: both members would pick each other as nearest neighbor with cosine ~ 1 , dragging the whole average up even when the rest of the text is disconnected. 2-NN demands that every idea has **multiple** close relatives, which is what “coherent” actually means. Grid search over 720 configurations confirms 2-NN is the only choice that distinguishes coherent multi-part passages from bags of unrelated facts.

The answer to: “*How connected are the ideas to each other?*”

3.4 Why three coordinates, not one

Consider five genres profiled on `all-mpnet-base-v2`:

Genre	N	B	I
Poetry (Dickinson)	3.3	0.3	0.58
arXiv abstracts	3.4	1.0	0.50
EUR-Lex legal	3.2	2.3	0.52
Dialogue	2.9	2.7	0.24
Code comments	3.2	1.4	0.16

Poetry: narrow and coherent. Legal: broad and coherent. Dialogue: broad and incoherent. Code comments: disconnected fragments. No single scalar separates these — the three-vector does.

4. The axiomatic foundation: the SIL theorem

Before introducing the three-coordinate profile, the paper asks: *if you demand a single scalar I : $\text{Text} \rightarrow \mathbb{R}$ with natural properties, what form must it take?*

4.1 The six axioms

Within a fixed frame (E, μ_0) — i.e., with the embedding model and baseline held constant — require:

1. **Paraphrase invariance.** If two texts land at (approximately) the same point in embedding space, they get the same I . *Outsources the meaning-equality judgment to E .*
2. **Redundancy non-increase.** Duplicating a segment never increases I . *Meaning $\neq$ volume.*
3. **Novelty monotonicity.** For fixed covariance, I is monotone in $||\mu_T - \mu_0||$. *Farther from generic $\Rightarrow$ at least as much information.*
4. **Idea additivity.** For texts in orthogonal embedding subspaces, $I(T_1 \oplus T_2) = I(T_1) + I(T_2)$. *Semantic analogue of Shannon additivity; unrelated ideas combine additively.*
5. **Orthogonal invariance.** I is invariant under rotations of embedding space. *No privileged axes.*
6. **Continuity.** I is continuous in the embeddings. *Encoder noise should not cause discontinuous jumps.*

Notation $\oplus$ means text concatenation, asserted as additive specifically when the two texts live in orthogonal subspaces (so the combined covariance is block-diagonal — a geometric direct sum).

4.2 The uniqueness theorem (SIL)

Under axioms 1–6, any such measure has the form, up to a positive scale constant,

$$I_E(T) = \|\mu_T - \mu_0\| \cdot \text{rank}(C_T).$$

Proof sketch. - Axioms 3 and 5 reduce dependence on μ_T to the single scalar $S(T) = ||\mu_T - \mu_0||$. - Axiom 5 reduces dependence on C_T to its spectrum (eigenvalues). - Axiom 2 eliminates

eigenvalue magnitudes (duplication rescales them without adding new directions), leaving $\text{rank}(\mathbf{C}_T)$. - Axiom 4 imposes a Cauchy functional equation on the joint dependence; continuity (axiom 6) forces linear solutions. A second Cauchy equation yields the product form.

4.3 Why SIL isn't the final answer

The theorem is a **frame-conditional uniqueness result**, not a universal law. And empirically, the factor $\text{rank}(\mathbf{C}_T)$ **saturates** — in 768-dim space, every text with k segments scores $\text{rank} = k-1$, because encoder noise makes almost every eigenvalue strictly nonzero. The factor carries no signal. Meanwhile $||\mu_T - \mu_0||$ varies richly. So the product **dominates on displacement** alone, collapsing a nominally two-coordinate scalar into effectively one coordinate.

That inadequacy is the motivation for replacing SIL with the three-coordinate profile of §3, where rank becomes $\mathbf{D}_{\text{eff}}$ (smooth, informative) and more structural information enters via $\mathbf{R}$ and $\mathbf{I}$.

5. The semantic quantum

§2.8 introduced deduplication via a threshold τ . The paper elevates this to a **discrete-unit interpretation**.

5.1 Definition

Given text T with segment embeddings $\mathbf{X}_T$ and threshold τ , the **quantum set** is $\mathbf{Q}_\tau(T) = \tilde{T} = \{c_1, \dots, c_m\}$ — the centroids from agglomerative clustering at cosine threshold $1 - \tau$. Each c_j is a **semantic quantum** at resolution τ . The **quantum count** is $\mathbf{m}_\tau(T) = |\mathbf{Q}_\tau(T)|$.

5.2 Three regimes

- **Sub-quantum ($\mathbf{m}_\tau \leq 1$):** single centroid. Novelty defined, breadth = 0, integration conventionally maximal. The text is a point, not a structure.
- **Single-quantum ($\mathbf{m}_\tau = 2$):** two centroids. Minimal configuration for which all three coordinates are non-trivially defined.
- **Multi-quantum ($\mathbf{m}_\tau \geq 3$):** generic regime where breadth and integration vary independently.

5.3 τ as measurement resolution

Parallel to physical measurement: below scale $1 - \tau$, the apparatus cannot resolve separate semantic units. τ is the Planck-like resolution parameter of the framework. Default: - $\tau = 0.70$ for natural prose (literature, journalism). - $\tau = 0.90$ for adversarial near-duplicate stress tests.

This is a **representational commitment** — made explicit rather than hidden.

6. The no-go theorem: trade-off triangle

Once you have the profile $(\mathbf{N}, \mathbf{B}, \mathbf{I})$, how do you collapse it to a single number? The paper proves: **you can't do it well**. No scalar can satisfy more than one of three natural desiderata.

6.1 The scalar family

Consider every scalar of the form

$$S(T; \Phi) = \Phi(\varphi_N(N(T); \mathcal{R}), \varphi_B(B(T); \mathcal{R}), \varphi_I(I(T); \mathcal{R}))$$

where $\mathcal{R}$ is a reference set of profiles used for per-coordinate normalization. Two normalizations matter:

- **Min-max:** $\varphi(x) = (x - \min \mathcal{R}_X) / (\max \mathcal{R}_X - \min \mathcal{R}_X)$. Linear, Lipschitz, but compressed by outliers.
- **Rank:** $\varphi(x) = \text{rank}_{\mathcal{R}}(x) / |\mathcal{R}|$. Bounded and outlier-proof, but piecewise constant (discontinuous at every rank boundary).

6.2 The three desiderata

(A) Analytic stability. Paraphrases get close values (Lipschitz continuity), and $S(\tau_1 \oplus \tau_2)$ is a continuous function of $S(\tau_1)$, $S(\tau_2)$ (closed-form composition).

(O) Ordinal robustness. On a benchmark of ordinal discrimination tasks, the scalar ranks categories correctly under noise; bootstrap pass rate exceeds $0.5 + \delta_0$; Benjamini–Hochberg correction survives.

(R) Cross-representation comparability. Under different embeddings $\mathbf{E}_1$, $\mathbf{E}_2$ with $\text{CKA} \geq c$, the scalars agree at Spearman $\rho \geq g(c)$, using a **shared reference set** without per-embedding refitting.

6.3 The theorem

For any reasonable tolerances, no scalar in the family satisfies more than one of (A), (O), (R) simultaneously.

Sketch of the three exclusions.

- **(A) $\Rightarrow \neg(\text{O})$.** (A) requires Lipschitz φ . Among reference-normalized schemes, only min-max is Lipschitz. Min-max compresses interiors whenever $\mathcal{R}$ contains outliers — which it always does on a non-degenerate benchmark. Compression prevents ordinal discrimination. So (A) $\Rightarrow \neg(\text{O})$.
- **(O) $\Rightarrow \neg(\text{A})$.** (O) requires rank-based φ (invariant to monotonic bootstrap distortions). Rank is discontinuous $\Rightarrow$ fails Lipschitz. Rank also has no closed-form composition: $\text{rank}(\tau_1 \oplus \tau_2)$ depends on where the concatenation lands in $\mathcal{R}$, not on $\text{rank}(\tau_1)$ and $\text{rank}(\tau_2)$ alone. So (O) $\Rightarrow \neg(\text{A})$.
- **(R) $\Rightarrow \neg(\text{A})$, $\neg(\text{O})$.** Shared- $\mathcal{R}$ across embeddings forces non-identity re-scaling under different $\mathbf{E}$, breaking both Lipschitz and ordinal agreement on non-extreme items.

6.4 Empirical confirmation

Every scalar in the family — SIL, S_{minmax} , S_{rank} , single-coordinate scalars under either normalization — lands at **at most one corner** on the actual benchmark. No empirical violation.

6.5 What the theorem licenses

Three architectural commitments follow:

1. The **profile (N, B, I) is the primary object**. It is cross-representation stable coordinate-wise and preserves information a scalar must destroy.
 2. The framework must expose **multiple scalars**, each for a different use case. Not one, not three — **two**: one for (A), one for (O). Corner (R) is unreachable.
 3. The **choice of scalar is left to the user**. The framework refuses to pick for you, because no pick is universally right.
-

7. Two recommended scalars

Both share a weighted geometric mean form and differ only in normalization.

7.1 The shared shape

$$S(T) = (\tilde{N}^\alpha \cdot \tilde{B}^\beta \cdot \tilde{I}^\gamma)^{1/(\alpha+\beta+\gamma)}$$

- **Geometric mean, not arithmetic**. Any factor going to zero drives S to zero. Prevents one big coordinate from masking a failed one.
- **Normalized exponent** $1/(\alpha+\beta+\gamma)$ keeps the scale comparable regardless of weight magnitude.
- **Weights** $(\alpha, \beta, \gamma) = (0.5, 3.0, 1.0)$ — set by grid search over 720 configurations on the synthetic benchmark. Breadth dominates; novelty saturates quickly and gets the smallest weight.

7.2 S_minmax — for analytic work

Normalization: $\varphi(x) = (x - \min R) / (\max R - \min R) \in [0, 1]$.

Lives at corner **(A)**: Lipschitz $\Rightarrow$ bounded paraphrase drift, compositionality for concatenation.
Use for: - Summarization-loss decomposition $\Delta S = \Delta N + \Delta B + \Delta I$ with interpretable magnitudes.
- Theoretical paraphrase-drift bounds. - Reasoning about $S(T_1 \oplus T_2)$ from $S(T_1)$, $S(T_2)$.

Price: outlier compression $\Rightarrow$ fails ordinal discrimination.

7.3 S_rank — for ranking work

Normalization: $\varphi(x) = \varepsilon + (1 - \varepsilon) \cdot \text{pct_rank}(x; R) \in (\varepsilon, 1]$ with $\varepsilon = 0.05$ floor (so the geometric mean never collapses to zero).

Lives at corner **(O)**: rank is invariant to monotonic bootstrap perturbation $\Rightarrow$ strong ordinal discrimination. Use for: - Leaderboards, document ranking, ordinal benchmarks. - Comparing texts of very different lengths (immune to outlier influence on R).

Price: discontinuous at rank boundaries $\Rightarrow \sim 2\times$ the paraphrase drift of S_minmax, and no closed-form composition.

7.4 The choice is structural, not tunable

You **cannot** combine min-max and rank to get both corners. The no-go theorem says so. Pick the scalar that matches your use case.

8. Empirical validation — scope and shape

All numbers live in `validation.ipynb`; this section describes what the notebook tests, not specific values.

8.1 Stability

- **Triplication drift:** concatenate each benchmark item to itself three times; measure how much the scalar moves. Should be ~ 0 (axiom 2). Is.
- **Paraphrase drift:** hand-constructed paraphrase pairs; point estimates plus bootstrap CIs. `S_minmax` is stable; `S_rank` less so (rank boundaries).

8.2 Ordinal benchmark — 28 checks

28 pairwise discrimination tasks (e.g., “coherent multi-part passage > bag of unrelated facts”, “multi-idea > single-idea”). Report: - Point-estimate pass rate. - Bootstrap mean pass rate over 300 resamples. - Benjamini–Hochberg-corrected significance at $\alpha = 0.05$.

`S_rank` dominates; `S_minmax` is substantially behind; single-coordinate scalars trail further.

8.3 Baselines comparison

Seven baselines run through the same 28-check protocol: type-token ratio, unigram entropy, mean sentence length, raw Euclidean novelty, mean-embedding norm, BERTScore-F, breadth alone. The framework’s `S_rank` outperforms all of them.

8.4 Cross-model robustness

Three embedding models (`all-mpnet-base-v2`, `all-MiniLM-L6-v2`, `all-distilroberta-v1`). Report per-coordinate Spearman ρ across models and scalar-level ρ . **Coordinates are more stable across encoders than any scalar summary** — consistent with the profile-first architecture.

8.5 External corpora

- **Project Gutenberg:** 5 novels, 507 chapters. Natural-text validation, long-form.
- **External short-form:** STS-B, SICK (sentence similarity), arXiv abstracts, EUR-Lex legal, poetry, dialogue, code comments. Each genre gets a (**N**, **B**, **I**) signature matching intuition.
- **Disclaimer for STS-B/SICK:** single-sentence items collapse the profile to raw cosine similarity. These numbers measure the **encoder**, not the framework.

8.6 SummEval (summarization-loss validation)

100 CNN/DailyMail source–summary pairs with 4 human axes (coherence, consistency, fluency, relevance). Report Spearman ρ between humans and `cos_dir`, ΔN , ΔB , ΔI .

Honest outcome. `cos_dir` (directional fidelity) is the only measure that correlates meaningfully with human ratings (moderate ρ on relevance and coherence). The per-coordinate deltas, which worked on synthetic faithful/partial/lossy/off-topic labels, **do not transfer** to human judgments of machine summaries. Real summarizers move multiple coordinates simultaneously; human raters don't decompose coordinate-wise. Conclusion: for summarization diagnostics against humans, use `cos_dir` as the primary signal.

9. Variational characterization of breadth

Recall that $\mathbf{B} = \mathbf{D}_{\text{eff}} \cdot \mathbf{R}$ was introduced as a geometric recipe. §9 shows it is actually the solution to a clean optimization problem — promoting it from heuristic to derived quantity.

9.1 Determinants, briefly

For an $n \times n$ matrix $\mathbf{A}$, $\det(\mathbf{A})$ measures the **volume of the parallelepiped spanned by its rows**. - $\det = 0 \Rightarrow$ rows are linearly dependent $\Rightarrow$ parallelepiped is flat. - $|\det|$ large $\Rightarrow$ rows point in very different directions $\Rightarrow$ large volume. - $|\det|$ small but nonzero $\Rightarrow$ rows nearly parallel $\Rightarrow$ thin sliver.

9.2 Determinantal Point Processes (DPPs)

A DPP is a distribution over subsets $S \subseteq \{1, \dots, n\}$ where $P(S) \propto \det(\mathbf{L}_S)$. The restricted matrix $\mathbf{L}_S$ has large determinant exactly when the selected items are **both high-quality and mutually diverse** — duplicates collapse rows, low-quality items shorten them.

DPP-MAP (Maximum A Posteriori): pick the subset with highest probability, i.e., maximize $\log \det \mathbf{L}_S$.

9.3 The kernel $\mathbf{L}$ for this paper

$$L_{ij} = q_i q_j \max(0, \langle x_i, x_j \rangle)$$

- **Quality** $q_i = \exp(-\|x_i - \mu_0\|_M / \sigma)$ — per-segment novelty weight (Mahalanobis distance from baseline, z-scored across segments).
- **Similarity** — cosine, clipped at zero for PSD-ness.

Maximizing $\log \det \mathbf{L}_S$ simultaneously rewards novel segments (via $\mathbf{q}$) and discourages redundant pairs (via cosine similarity).

9.4 The objective and its solution

$$S^*(T) = \arg \max_{\emptyset \neq S \subseteq [n]} \log \det L_S.$$

- **log det is submodular.** Greedy selection (add the segment with highest marginal gain, stop when gain turns non-positive) gives the standard $(1 - 1/e)$ approximation guarantee. **On this paper's benchmark, greedy is in fact ILP-optimal** — matching exhaustive enumeration up to $|S| \leq 16$.
- The **stopping rule selects $|S|$ automatically**. No need to specify “pick 3 sentences” externally.

9.5 The decomposition theorem

$$\log \det L_S = 2 \sum_{i \in S} \log q_i + \log \det G_S,$$

where $G_S = [\langle x_i, x_j \rangle]$ is the cosine Gram matrix of S . This is an algebraic identity (pull quality diagonals out of the determinant). It splits the objective into a **quality term** (sum of log-qualities, essentially novelty) and a **pure diversity term** ($\log \det G_S$).

9.6 The correspondence with breadth

Under a near-isotropic regime (pairwise cosines $\approx$ some constant c within tolerance ε):

- The spectrum of G_S concentrates: one eigenvalue near $1 + (|S|-1)c$, the rest near $1 - c$.
- $D_{\text{eff}}(G_S) \approx |S|$ up to $O(\varepsilon) + O(|S| c^2)$ corrections.
- $R(S)$ is monotone increasing in $|S|$ and decreasing in c .
- Both $\log \det G_S$ and $B_S = D_{\text{eff}} \cdot R$ are **strictly increasing in $|S|$** and **strictly decreasing in c** — they co-vary.

Defining the parameter-free quantity

$$V(T) := \log \det L_{S^*(T)} - 2 \sum_{i \in S^*(T)} \log q_i$$

(the pure diversity contribution after stripping off quality), the corollary is: **V recovers B up to a monotonic transformation and $O(\varepsilon)$ error.** Empirical confirmation: Spearman $\rho(V, B) \approx 0.985$ across 507 Gutenberg chapters (number in notebook) — near-perfect monotonic agreement on natural text.

9.7 Free consequences

- **Order invariance (0% drift).** DPP objective is over unordered sets; shuffling sentences doesn't move V .
- **Triplication invariance (0% drift).** Duplicated segments produce linearly dependent rows in L ; determinant drops to zero contribution. Axiom 2 is enforced *structurally*, no deduplication step needed.
- **Parameter-free.** No weights (α, β, γ) , no trade-off λ . The geometry of volume does the balancing.

9.8 Why this matters

Before §9, $B = D_{\text{eff}} \cdot R$ was an engineering choice. A skeptical reader could ask: *why that combination?* Answer (before): “it works on the benchmark.” Answer (after): **it is (up to near-isotropic error) the log-volume of a diversity-maximizing DPP-MAP subset — the answer to a natural optimization question with nothing to do with the benchmark.**

The formula was not chosen — it was discovered.

10. Downstream: extractive summarization

The variational form is not just theoretically satisfying — it also works as a practical summarizer.

10.1 Setup

For each article, select sentences by greedy DPP log-det maximization; evaluate via ROUGE against the human reference. Baselines: Lead-3 (first three sentences), MMR (classical diversity selector with tuned λ), centroid selection, uniform random. Tested on XSum (186 articles).

10.2 MMR — the classical baseline

MMR (Maximal Marginal Relevance). Greedy selection under:

$$\text{MMR}(x) = \lambda \cdot \text{sim}(x, q) - (1 - \lambda) \cdot \max_{x' \in S} \text{sim}(x, x')$$

— a tunable trade-off between relevance and redundancy. Standard baseline for diverse extractive selection. Uses $\lambda = 0.5$ in the experiments.

10.3 Result

- **At DPP’s own (variationally-selected) size**, DPP significantly beats MMR on ROUGE-L, without any tunable hyperparameter. The parameter-free size selection is itself part of DPP’s contribution.
- **Forced to $k = 3$** , DPP and MMR are statistically tied. Lead-3 beats both — a known artifact of news corpora (lead bias), not a framework failure.

Interpretation. DPP’s empirical advantage over MMR appears precisely at the subset size the DPP objective itself prefers. Fixing the size externally removes that advantage. This is consistent with the theorem in §9.6: the log-det objective jointly balances $|S|$ and pairwise similarity, and fixing one of them breaks the balance.

11. Architecture summary

Three layers of the theory, in decreasing theoretical status:

Layer	Role	Status
SIL axioms + theorem (§4)	Six axioms $\rightarrow$ unique form $ \mu_{\top} - \mu_{\emptyset} \cdot \text{rank}(\mathbf{C}_{\top})$ within a frame.	Uniqueness within a frame; empirically inadequate (rank saturates); scope-setting, not deliverable.
Profile ($\mathbf{N}$, $\mathbf{B}$, $\mathbf{I}$) (§3)	Three orthogonal geometric coordinates.	Primary output of the framework. Cross-representation stable. Passes ordinal tests.

Layer	Role	Status
Two scalars S_{minmax}, S_{rank} (§7)	Use-case-specific summaries.	Practical conveniences; each picks a single corner of the no-go triangle.

Why this architecture and not a different one:

1. **One universal scalar is impossible** (no-go theorem, §6).
2. **One universal profile is possible** and captures what a scalar must destroy.
3. **Multiple scalars, chosen per use case**, is the correct response to an impossibility theorem: give the user exactly as many tools as the theorem permits useful ones.

12. Scope, caveats, and honest limitations

12.1 Representation-dependence is a feature

Every quantity in the framework depends on (E , μ_0 , τ , segmentation, reference set). This is **not** a defect to be engineered away. Three defenses:

1. **Inspectable commitments.** A representation-free measure would still make these choices — just hidden ones. Making E explicit turns hidden assumptions into auditable ones.
2. **Calibratable per domain.** Legal $E \neq$ biomedical $E \neq$ general-prose E . Exposing the choice lets practitioners pick what fits.
3. **Theorem-mandated.** The no-go theorem proves corner (R) is unreachable. Representation-free comparability isn't a missing feature — it's provably unavailable.

12.2 The SummEval result is mixed

Per-coordinate deltas (ΔN , ΔB , ΔI) correlate near-zero with human ratings on machine-generated summaries. The framework's attrition-signature story, as originally envisioned, **does not validate** against human judgment. What survives: `cos_dir` is a competitive unsupervised summarization-quality signal.

12.3 The 28-check benchmark is synthetic

The ordinal benchmark is hand-constructed. Definitive validation requires human ratings of semantic richness/coherence/informativeness, not yet done at scale. Some benchmark items (e.g., `single_backprop`) sit awkwardly in embedding space and cause specific failures that are encoder artifacts, not framework failures — but this distinction only becomes clear on close inspection.

12.4 Quantum lifetime is non-discriminative on short items

The time-resolved extension (persistence of centroids across sliding windows) is included in the framework but is not discriminative on 5-sentence benchmark items. It is only useful on long-form texts like book chapters.

13. Glossary (one-line references)

- **E** — sentence-embedding model.
 - **T**, **T_i**, **e_i** — text, its segments, their embeddings.
 - **n**, **k**, **m** — embedding dimension, raw segment count, deduplicated segment count.
 - **X_T**, **μ_T**, **C_T** — raw embedding cloud, its mean, its covariance.
 - **$\tilde{T}$** , **c_j** — deduplicated centroid set, an individual centroid.
 - **μ₀**, **Σ₀** — baseline mean, baseline covariance.
 - **||·||_M** — Mahalanobis norm.
 - **τ** — deduplication / measurement-resolution threshold.
 - **N**, **B**, **I** — novelty, breadth, integration coordinates.
 - **S_M** — Mahalanobis displacement from baseline (pre-log novelty).
 - **D_{eff}** — effective rank (exp of spectral entropy).
 - **R** — mean radial cosine distance from centroid mean.
 - **I_{2-NN}** — mean second-nearest-neighbor cosine similarity among centroids.
 - **S_{minmax}**, **S_{rank}** — the two recommended scalars.
 - **(α, β, γ)** — profile weights, fixed at **(0.5, 3.0, 1.0)**.
 - **ε** — rank-normalization floor, fixed at **0.05**.
 - **cos_{dir}** — $\cos(\mu_T, \mu_{\{T'\}})$; directional fidelity of a summary.
 - **ΔN**, **ΔB**, **ΔI**, **ΔS** — coordinate and scalar attrition of a summary.
 - **Q_τ(T)**, **m_τ** — quantum set at resolution **τ**, quantum count.
 - **L**, **G_S**, **q_i** — DPP kernel, Gram submatrix, per-segment quality.
 - **V(T)** — DPP log-volume (pure diversity contribution), recovers **B**.
 - **MMR** — Maximal Marginal Relevance; classical tunable diversity baseline.
 - **DPP-MAP** — most-probable subset under a Determinantal Point Process.
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14. Further reading (paper references)

- **geometric_profile_semantic_information.tex / .pdf** — the paper itself; contains the formal statement of all laws, measures, and theorems.
- **validation.ipynb** — every empirical number, reproducible.